

Source data, cleaning, and preparation

Original SQL

Math courses taken by first time freshmen were queried as follows. This data set is named “mathCourses” in the code below.

```
SELECT *
FROM OBIA.COMBINED_COURSE_V
WHERE (
    UPPER(DEPARTMENT) LIKE '%MATH%'
    OR UPPER(TITLE) LIKE '%MATH%'
    OR UPPER(SUBJECT_LONG) LIKE '%MATH%'
    OR UPPER(SUBJECT) LIKE '%MATH%'
)
AND TERMEXTRACT IN ('E','S')
AND EMPLID IN (
    SELECT EMPLID
    FROM OBIA.FTF_DEMO_V
);
```

Data describing first time freshmen were queried as follows. This data set is named “ftfData” in the code below.

```
SELECT * FROM OBIA.FTF_DEMO_V
```

The number of credits taken per student per term were queried as follows. This data set is named “creditLoad” in the code below.

```

SELECT
  TERM,
  EMPLID,
  SUM(UNITS) AS UNITS
FROM
  OBIA.COMBINED_COURSE_V
WHERE
  TERMEXTRACT IN ('E','S')
  AND EMPLID IN (
    SELECT EMPLID
    FROM OBIA.FTF_DEMO_V
  )
GROUP BY
  TERM,
  EMPLID

```

Preparation Description

The first term each student took a math course was identified, and all math classes taken in that term are extracted.

These initial math courses, student demographics (e.g. age and gender) and indicators of academic ability (e.g. ACT test scores and high school GPA), and the number of credit units per student per term are combined.

Some new fields were defined. A year difference (or “*yr_diff*”) was calculated as the difference between the cohort date and the end-of-term date in fractions of a year. Medians and standard deviations for high school GPA and ACT math test scores per course per year for successful students who achieved a GPA greater than 2.4 (or at least a grade of B-minus) were calculated.

A new feature was engineered. The distance (or “*dist*”) between a student’s high school GPA and ACT math test scores and the prior year’s median qualifications for successful students was calculated as a Euclidean distance using the median-centered z-score. (Using principal components in order to de-correlate the variables was investigated and not found beneficial.)

A variety of filters and transformations were applied as described below. Most notably, only complete cases were used in modeling, or all records missing a value were removed. This excluded the majority of students in recent years, as the rate of submitting ACT math test scores has precipitously declined.

Preparation Detail

New fields were defined:

- “*class*” was defined as a combination of the fields TERM, CATNBR (catalog number), and SECTION.
- “*course*” was defined as a combination of the fields SUBJECT_CD (subject code) and CATNBR (catalog number).
- “*class_yr*” (class year) was defined as the year extracted from the field “EOTDATE” (end-of-term date).
- “*yr_diff*” (year difference) was defined as the difference between the fields “COHORT_DT” (cohort date) and “EOTDATE” (end-of-term date) in fractions of years.
- *season* was defined as “spring”, “summer”, or “fall” according to the identifier numbers 4, 6, or 8 in the “TERM” field.
- *course_level* was extracted as the “CATNBR” (catalog number) divided by 1000.

One field was re-named:

- “*load*” replaced the term “UNITS” referring to the number of credits taken per student per term.

Filters were applied:

- All math labs are filtered out (where “GRADE” is blank or has values of NA and “UNITS” is equal to zero.)
- All math classes taken after the first term per student that math classes are taken are filtered out.
- All courses with CATNBR values of 3000 and above are filtered out.
- “Withdraw” grades (“W”) are removed.
- Courses that were not taught after the year 2021 were removed.
- 37 students with multiple cohort dates were removed.
- All records with missing or ‘NA’ values were removed (only complete cases were used.)

Outliers were removed:

- HSGPA values of zero (no high school GPA)

- Students aged less than eleven are filtered out.
- Unusually high credit load values are removed (greater than the 99th percentile)
- Early math classes are removed ($\text{yr_diff} < -0.5$, suggesting the class was taken during high school)
- Missing, rare or unusual grade values for this data set are removed (" ", "V", "I", "NC", "CR" and NA)

Some field values were transformed:

- "APCREDIT" (advanced placement credit) values of NA are converted to zero.
- "GRADE" values of "EU" are converted to zero.

Data sets were joined:

- Selected fields describing first time freshmen were merged with the initial math courses data
- The number of credits ("UNITS" re-named as "load") were merged into this data set.

New features were engineered:

- The median and standard deviation values for high school GPA and ACT math test scores were calculated per course per year for students who achieved a grade higher than 2.4 (in other words, at least a B-minus.)
- The z-scores for high school GPA and ACT math test scores were calculated using the prior year's median and standard deviation for these 'successful' students.
- A new field "*dist*" was calculated as the Euclidean distance in these median-centered z-scores. It is the square root of the squared ACT math difference plus the squared high school GPA difference.

Variables Described

Demographics:

- Age (bucketed as less than 17 yrs old/17/18/19 or 20/greater than 21 years old)
- Sex (male or female)
- First generation status
- Ethnicity

High school:

- High school GPA
- Number of AP credits

- Whether the high school was public or private

Test scores:

- ACT Math
- ACT English
- ACT Science
- ACT Comprehensive

University information:

- Math course taken
- Course level (as 1000 or 2000)
- Residence status
- Pell grant eligibility
- Cohort year (2005 to 2024)
- Class year (2005 to 2025)
- Years from the cohort date (always fall of the year) until the first term with math courses (as yr_diff)
- Season (spring, summer, or fall)
- Honors status

Derived features:

- The high school GPA converted to a z-score using the prior year's course median successful student
- The ACT math test scores converted to a z-score using the prior year's course median successful student
- The combined Euclidean distance of these z-scores

Grade Mapping

Table 1: Grade Mapping

Letter Grade	GPA
E, EU	0.0
D-	0.7

Letter Grade	GPA
D	1.0
D+	1.3
C-	1.7
C	2.0
C+	2.3
B-	2.7
B	3.0
B+	3.3
A-	3.7
A	4.0

Code Detail

```
#####
## LOAD ##
#####

ftfData <- readRDS(here::here("Data", "Freshman_data.rds"))
mathCourses <- readRDS(here::here("Data", "FTF_math_data.rds"))
creditLoad <- readRDS(here::here("Data", "Freshman_career_credit_load.rds" ))

#####
## PREP ##
#####

# convert to numeric
mathCourses$CATNBR <- as.numeric(as.character(mathCourses$CATNBR))

# identify unique classes
mathCourses$class <- paste(mathCourses$TERM, mathCourses$SUBJECT_CD, mathCourses$CATNBR, mathCourses$SECTID)

# identify unique courses
mathCourses$course <- paste(mathCourses$SUBJECT_CD, mathCourses$CATNBR, sep="_")

# re-arrange order
mathCourses <- mathCourses[,c("TERM", "course", "class", "CATNBR", "EMPLID", "TITLE", "SECTION")]

## IDENTIFY FIRST TERM OF MATH CLASSES ##
```

```

# Identify minimum term per student
minTerm <- aggregate(TERM ~ EMPLID, data = mathCourses, min, na.rm = TRUE)
minTerm$reference <- paste(minTerm$EMPLID, minTerm$TERM, sep = "_") # combine for reference
mathCourses$minTerm <- paste(mathCourses$EMPLID, mathCourses$TERM, sep = "_") %in% minTerm$reference

firstTermFilter <- mathCourses$minTerm == TRUE

mathLabFilter <-
  mathCourses$GRADE %in% c(" ", NA) &
  mathCourses$UNITS == 0

firstCourses <- mathCourses[firstTermFilter & !mathLabFilter,]

# Remove students with multiple values for COHORT_DT in FTF data
# problemIDs <- c("00517160", "00572252", "00280932", "00643029", "00548206")

theDupes <- table(ftfData$EMPLID) |>
  (\\(x){x[x>1]}()) |>
  names()

duplicateFilter <- ftfData$EMPLID %in% theDupes

# Merge in FTF data
student_demographics <- c("SEX",
  "FIRST_GEN_STATUS_CD",
  "ETHNICITY",
  "RESSTAT",
  "FA_PELL",
  "AGE"
)

academic_prep <- c(
  "APCREDIT",
  "HSGPA",
  "HSPRIVATE",
  "HONORS",
  "ACTCOMP",
  "ACTENGL",
  "ACTMATH",
  "ACTSCI" #,
  # SATMATH, # leaving out SAT because it's a smaller proportion
  # SATVERBAL,
  # SATWRITG

```

```

)

firstCourses <- merge(firstCourses, ftfData[, c("EMPLID","COHORT_DT", student_demographics,

# Merge in credit load
names(creditLoad)[names(creditLoad) == "UNITS"] <- "load"

firstCourses <- merge(firstCourses, creditLoad, by = c("TERM", "EMPLID"), all.x = TRUE)

# Freshman year
firstCourses$cohort_year <- year(firstCourses$COHORT_DT)

# Class year
firstCourses$class_year <- year(firstCourses$EOTDATE)

# Year to class
firstCourses$yr_diff <- time_length(interval(firstCourses$COHORT_DT, firstCourses$EOTDATE),

# Season

# Align season
season_map <- c("4" = "spring", "6" = "summer", "8" = "fall")

# Extract 4th digit and map
firstCourses$season <- season_map[substr(firstCourses$TERM, 4, 4)]

# Course levels column
firstCourses$course_level <- firstCourses$CATNBR %/% 1000

# filter out courses at 3000+ level
course_levelFilter <- firstCourses$course_level < 3

# filter out courses that aren't recent
course_by_year <- table(firstCourses[course_levelFilter, c("course","class_year") ]) |> as.
recentCourses <- unique(course_by_year$course[course_by_year$class_year %in% c(2021:2025) &
recent_filter <- firstCourses$course %in% recentCourses

# Age buckets
firstCourses$age_cut <- cut(firstCourses$AGE, breaks = c(0,17,18,19,21,100))

# Age filter
ageFilter <- firstCourses$AGE > 11

```



```
print("Prep complete")
```

```
[1] "Prep complete"
```

```
## GRADE TRANSFORMATIONS ##
```

```
# convert grade value names to values compatible with R naming
```

```
clean_grade <- function(x) {  
  x <- as.character(x)  
  x <- gsub("-", "_minus", x) # replace '-' with '_minus'  
  x <- gsub("\\+", "_plus", x) # replace '+' with '_plus'  
  x <- gsub(" ", "missing", x) # replace spaces with '_'  
  
  return(x)  
}  
  
firstCourses$cleanGrade <- clean_grade(firstCourses$GRADE)  
  
firstCourses$cleanGrade <- factor(firstCourses$cleanGrade,  
                                levels = c(  
                                  "A", "A_minus", "B_plus", "B", "B_minus", "C_plus",  
                                  "C", "C_minus", "D_plus", "D", "D_minus",  
                                  "E", "EU", "W", "I", "CR", "NC", "V",  
                                  "missing", NA  
                                )  
)
```

```
# RARE AND UNUSUAL VALUES
```

```
rareGrades <- firstCourses$cleanGrade %in% c("missing", "V", "I", "NC", "CR") # , "EU" conver
```

```
# WITHDRAW
```

```
firstCourses$wdraw_binary <- NA
```

```
firstCourses$wdraw_binary <- ifelse(firstCourses$GRADE == "W", "withdraw", "not_withdraw")
```

```
## CONVERT APCREDIT NA VALUES to ZERO ##
```

```
firstCourses$APCREDIT[is.na(firstCourses$APCREDIT)] <- 0
```

```
print("Transformations complete")
```

```
[1] "Transformations complete"
```

```
#####  
## PRE-PROCESS ##  
#####  
  
# Select columns  
  
# colnames(popCourses)  
  
keepColumns <- c(  
  "EMPLID" ,  
  "course" ,  
  "SEX" ,  
  "FIRST_GEN_STATUS_CD" ,  
  "ETHNICITY" ,  
  "RESSTAT" ,  
  "FA_PELL" ,  
  "APCREDIT" ,  
  "HSGPA" ,  
  "HSPRIVATE" ,  
  "HONORS" ,  
  "ACTCOMP" ,  
  "ACTENGL" ,  
  "ACTMATH" ,  
  "ACTSCI" ,  
  "cohort_year" ,  
  "class_year" ,  
  "yr_diff" ,  
  "season" ,  
  "course_level" ,  
  "age_cut" ,  
  "cleanGrade" ,  
  "HSGPA.z" ,  
  "ACTMATH.z" ,  
  "dist"  
)  
  
target_classes <- c("GRADEGPA")  
  
#####
```

```

## CLEAN ##
#####

cleanFilters <- !is.na(firstCourses$cleanGrade) &
  !rareGrades
#!popCourses$cleanGrade %in% c("missing", "V", "I", "NC", "CR")

# Outlier filters
zeroHSGPA_filter <- firstCourses$HSGPA == 0
highLoad_filter <- firstCourses$load > quantile(firstCourses$load, 0.99)
preMath_filter <- firstCourses$yr_diff < -0.5 # since all cohort dates start in September, I
outlierFilters <- !zeroHSGPA_filter & !highLoad_filter & !preMath_filter

# withdrawFilter
withdrawFilter <- firstCourses$cleanGrade != "W"

# Combined filters
cleanData <- firstCourses[cleanFilters & outlierFilters & course_levelFilter & recent_filter
cleanData <- cleanData[!is.na(cleanData$cleanGrade),]
cleanData$cleanGrade <- droplevels(cleanData$cleanGrade)

cleanData$course <- factor(cleanData$course)
cleanData$ETHNICITY <- factor(cleanData$ETHNICITY)

#####
## DISTANCE FROM MEDIAN SUCCESSFUL STUDENT ##
#####

hiGrades <- aggregate(cbind(HSGPA,ACTMATH) ~ course + class_year,
  data = cleanData[cleanData$GRADEGPA > 2.4,], function(x){
    c(median = median(x),
      IQR = IQR(x),
      stdev = sd(x),
      Q = quantile(x, probs = c(0.1,0.25,0.5,0.75,0.9), na.rm = TRUE),
      count = length(x)
    )
  }) |>

  (\(x){
    do.call(data.frame,x)
  })()

```

```

# Merge to cleanData

hiGrades$class_year_shift <- hiGrades$class_year + 1

cleanData <- merge(cleanData, hiGrades[,c("course", "class_year", "class_year_shift", "ACTMATH.z", "HSGPA.z")], by="course", all=TRUE)

# Calculate distance as a z-score

cleanData$ACTMATH.z <- (cleanData$ACTMATH - cleanData$ACTMATH.median)/cleanData$ACTMATH.stdev
cleanData$HSGPA.z <- (cleanData$HSGPA - cleanData$HSGPA.median)/cleanData$HSGPA.stdev

cleanData$dist <- sqrt(
  ((cleanData$ACTMATH - cleanData$ACTMATH.median)/cleanData$ACTMATH.stdev)^2 +
  ((cleanData$HSGPA - cleanData$HSGPA.median)/cleanData$HSGPA.stdev)^2
)

```

Data Detail

```

cleanData[, c(target_classes, keepColumns)] |>
  (\(x){x[complete.cases(x),]})( ) |>
  skim()

```

Table 2: Data summary

Name	(function(x) {
x[comp...	
Number of rows	38920
Number of columns	26
Column type frequency:	
character	6
factor	4
numeric	16
Group variables	None

Variable type: character

skim_variable	n_miss- ing	com- plete_rate	min	max	empty	n_unique	whitespace
EMPLID	0	1	8	8	0	38601	0
SEX	0	1	1	1	0	2	0
FIRST_GEN_STA- TUS_CD	0	1	1	1	0	3	0
RESSTAT	0	1	1	1	0	2	0
HSPRIVATE	0	1	1	1	0	2	0
season	0	1	4	6	0	3	0

Variable type: factor

skim_vari- able	n_miss- ing	com- plete_rate	or- dered	n_unique	top_counts
course	0	1	FALSE	27	MAT: 8911, MAT: 6327, MAT: 5140, MAT: 3480
ETHNIC- ITY	0	1	FALSE	9	C: 27934, H: 4843, A: 2802, M: 1864
age_cut	0	1	FALSE	5	(17: 31732, (18: 3341, (19: 1949, (0,: 1594
cleanGrade	0	1	FALSE	13	A: 11672, B: 4576, A_m: 4333, B_p: 3746

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
GRADEGPA	0	1	2.86	1.22	0.00	2.30	3.30	4.00	4.00	
FA_PELL	0	1	0.22	0.41	0.00	0.00	0.00	0.00	1.00	
AP- CREDIT	0	1	7.32	12.43	0.00	0.00	0.00	12.00	85.00	
HSGPA	0	1	3.63	0.34	1.07	3.42	3.72	3.92	4.42	
HONORS	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
ACT- COMP	0	1	25.08	4.36	10.00	22.00	25.00	28.00	36.00	
ACTENGL	0	1	24.86	5.31	7.00	21.00	24.00	29.00	36.00	
ACT- MATH	0	1	24.45	4.64	1.00	21.00	24.00	27.00	36.00	
ACTSCI	0	1	24.98	4.57	2.00	22.00	24.00	28.00	36.00	

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
co- hort_year	0	1	2015.06	5.34	2005.00	2011.00	2016.00	2019.00	2024.00	
class_year	0	1	2015.42	5.26	2006.00	2011.00	2016.00	2020.00	2025.00	
yr_diff	0	1	0.51	0.80	-0.40	0.24	0.26	0.26	16.64	
course_level	0	1	1.03	0.18	1.00	1.00	1.00	1.00	2.00	
HSGPA.z	0	1	-0.54	1.24	-	-1.16	-0.23	0.32	2.01	
					42.07					
ACT- MATH.z	0	1	-0.14	1.12	-	-0.81	0.00	0.51	6.93	
					10.00					
dist	0	1	1.42	1.04	0.00	0.72	1.18	1.87	42.07	